DEPARTMENT OF STATISTICS

Udai Pratap Autonomous College

Varanasi

MSc STATISTICS: SECOND SEMESTER

PROJECT REPORT

Factors Affecting

Life Expectancy Across Nations

WHO Global Health Data (2000–2015)

Submitted by

Om Upadhyay

Roll No: 2524003

Enrolment No: STA/25/003

Under the Supervision of

Prof. Narendra Pratap Singh

Head of Department, Department of Statistics

Session: 2025–2026

CERTIFICATE

This is to certify that the project report entitled "Factors Affecting Life Expectancy Across Nations" has been prepared and submitted by Om Upadhyay, Roll No. 2524003, in partial fulfilment of the requirements for the degree of Master of Science in Statistics, Udai Pratap Autonomous College, Mahatma Gandhi Kashi Vidyapith, for the academic session 2025–2026.

The project work has been carried out under my supervision and guidance. To the best of my knowledge, the work presented in this report is original, has not been submitted elsewhere for any other degree or diploma, and all sources of information used have been duly acknowledged.

I recommend this project report for acceptance and evaluation.

Project Supervisor	Head of Department
Prof. Narendra Pratap Singh	Prof. Narendra Pratap Singh
Head of Department	Department of Statistics
Udai Pratap Autonomous College	Udai Pratap Autonomous College
Date: _____	Date: _____

DECLARATION

I, Om Upadhyay, a student of MSc Statistics (Second Semester), hereby declare that the project report entitled "Factors Affecting Life Expectancy Across Nations" submitted to Udai Pratap Autonomous College, Mahatma Gandhi Kashi Vidyapith, is a record of original work carried out by me under the supervision of Prof. Narendra Pratap Singh.

I further declare that the work presented in this report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma. Wherever reference has been made to the work of others, it has been clearly acknowledged in the text and in the references section.

The dataset used in this study is sourced from the WHO Global Health Observatory Data Repository and is publicly available for academic use. The statistical analysis was conducted independently using Python programming language.

Name: Om Upadhyay

Roll No: 2524003

Date: _____

Place: Varanasi

ACKNOWLEDGEMENT

The completion of this project report would not have been possible without the support and guidance of several individuals, to whom I express my heartfelt gratitude.

I am deeply thankful to Prof. Narendra Pratap Singh, my project supervisor, for providing constant encouragement, constructive feedback, and academic direction throughout the course of this study. Your patience and expertise have shaped this work significantly.

I also extend my sincere gratitude to the Head of the Department of Statistics, Prof. Narendra Pratap Singh, for providing the necessary academic infrastructure and for encouraging students to engage with real-world datasets and statistical methodologies that go beyond the classroom.

I would like to acknowledge the World Health Organization (WHO) for making the Global Health Observatory dataset publicly available for academic research. This dataset, covering health and development indicators across 193 countries from 2000 to 2015, was the backbone of this study.

Finally, I thank my family for their unwavering support and patience throughout this academic endeavour.

Om Upadhyay

MSc Statistics, Second Semester

Udai Pratap Autonomous College

ABSTRACT

Life expectancy is one of the most comprehensive and widely studied indicators of a nation's overall health, development, and well-being. Understanding the factors that cause substantial variation in life expectancy across different countries is a question of significant academic and policy importance. This project undertakes a detailed empirical study of the socioeconomic, health, and environmental determinants of life expectancy using the World Health Organization's Global Health Observatory dataset, which covers 193 countries over the period 2000 to 2015.

The dataset, comprising 2,938 observations across 22 variables, was subjected to a multi-layered statistical analysis including descriptive statistics, Pearson correlation analysis, independent samples t-test, and multiple linear regression modelling. Descriptive analysis revealed that the global mean life expectancy over the study period was 69.22 years, with a standard deviation of 9.51 years, and the values ranging from a minimum of 36.30 years to a maximum of 89.00 years.

A statistically significant difference was found between developed and developing nations, with developed countries recording a mean life expectancy of 79.20 years compared to 67.12 years for developing countries: a gap of 12.08 years. The independent samples t-test confirmed this difference to be highly significant ($t = 29.8052$, $p < 0.001$), decisively rejecting the null hypothesis.

Correlation analysis identified schooling ($r = 0.7151$) as the strongest positive predictor and adult mortality ($r = -0.6964$) as the strongest negative correlate of life expectancy. Multiple linear regression modelling produced a model with an R-squared value of 0.7938, indicating that approximately 79.4% of the total variation in life expectancy across countries is explained by the selected predictors. Schooling emerged as the most significant independent predictor, with every additional year of average schooling associated with an increase of approximately 1.0064 years in life expectancy, holding all other variables constant.

These findings underscore the critical role of education, disease control, and healthcare investment in shaping population health outcomes across nations.

Keywords: Life Expectancy, WHO Dataset, Correlation Analysis, Multiple Linear Regression, Hypothesis Testing, Descriptive Statistics, Schooling, Adult Mortality, HIV/AIDS, Developing Nations.

TABLE OF CONTENTS

SECTION	PAGE
Certificate	ii
Declaration	iii
Acknowledgement	iv
Abstract	v
1. Introduction	1
2. Review of Literature	4
3. Objectives of the Study	7
4. Hypotheses	8
5. Methodology	9
6. Descriptive Statistics	12
7. Visual Analysis and Graphical Representation	17
8. Correlation Analysis	24
9. Hypothesis Testing	28
10. Multiple Linear Regression Analysis	32
11. Discussion of Findings	40
12. Conclusion	43
13. Limitations and Scope for Future Research	45
References	47
Appendix: Raw Data Sample (First 20 Observations)	49

1. INTRODUCTION

1.1 Background

The length and quality of human life have long been central concerns in public health, economics, and policy research. Life expectancy at birth, defined as the average number of years a newborn is expected to live if the prevailing patterns of mortality at the time of birth remain the same throughout life: is universally regarded as one of the most informative and widely used summary measures of a population's health status. It integrates the effects of mortality conditions across all age groups into a single, interpretable figure, making cross-country and cross-temporal comparisons both meaningful and revealing.

Despite significant improvements in global health over the past century, life expectancy remains highly unequal across nations. While citizens of Japan, Switzerland, and Australia routinely live into their mid-eighties, those born in some sub-Saharan African nations may expect to live for only fifty to sixty years. The magnitude of this disparity, sometimes exceeding thirty years between countries: points to profound structural differences in healthcare systems, economic development, education, nutrition, disease burden, and governance. Understanding these disparities is not merely an academic exercise; it carries enormous implications for public health policy, international development assistance, and global equity.

The World Health Organization (WHO) has been systematically collecting data on health indicators across its member states for several decades. The dataset used in this study is drawn from the WHO Global Health Observatory and covers 193 countries over the period 2000 to 2015. During this period, the global health landscape underwent significant changes: the Millennium Development Goals drove targeted investments in child health and infectious disease control, antiretroviral therapy transformed the prognosis for HIV-positive individuals, and expanding access to education began yielding visible demographic dividends in developing regions.

1.2 Relevance of the Study

Statistical analysis offers tools that go far beyond simple description. Through correlation analysis, hypothesis testing, and regression modelling, a statistician can determine which factors are most strongly associated with longevity, whether observed differences between groups are statistically real or attributable to chance, and which

predictors contribute most independently to explaining variation in health outcomes. These findings, grounded in mathematical reasoning, can guide evidence-based policymaking with a precision that intuition alone cannot provide.

This project applies precisely these tools to a globally relevant question: what are the most significant determinants of life expectancy across nations? By working with a large, real-world dataset of nearly 3,000 observations, the analysis moves beyond the limitations of small, single-country surveys and into the domain of robust, generalizable statistical inference. The 22 variables in the dataset span economic indicators such as gross domestic product (GDP) per capita, health system measures such as immunisation coverage and health expenditure, behavioural factors such as alcohol consumption, and disease burden indicators such as HIV/AIDS prevalence and adult mortality rates.

1.3 Statistical Methods Employed

The analysis proceeds in four stages. First, descriptive statistics, including measures of central tendency and dispersion: are computed to characterise the distribution of life expectancy and its predictors. Second, Pearson correlation coefficients are calculated to quantify the strength and direction of linear relationships between life expectancy and each predictor variable. Third, an independent samples t-test is conducted to formally test whether the observed difference in mean life expectancy between developed and developing countries is statistically significant. Fourth, a multiple linear regression model is estimated to identify the independent contribution of each predictor to life expectancy, controlling for the influence of other variables simultaneously.

Together, these methods provide a comprehensive statistical portrait of global life expectancy covering all four dimensions: descriptive, relational, inferential, and predictive.

1.4 Data Source

The dataset was obtained from Kaggle, where it is publicly available under the title 'Life Expectancy (WHO)' contributed by user Kumarajarshi. The original source of the data is the WHO Global Health Observatory Data Repository. The dataset contains health-related information for 193 countries from the year 2000 to 2015, resulting in a total of 2,938 observations across 22 variables, including both numerical and categorical data types.

2. REVIEW OF LITERATURE

The question of what determines life expectancy has attracted sustained scholarly attention across disciplines ranging from epidemiology and economics to sociology and public health. A substantial body of empirical research has examined this question using a variety of datasets, methodologies, and geographical scopes. The following review synthesises key findings from relevant studies and positions the present investigation within the existing literature.

2.1 Economic Development and Life Expectancy

Preston (1975) documented what became known as the 'Preston Curve', a positive, concave relationship between national per capita income and life expectancy. His analysis demonstrated that wealthier nations tended to have longer life expectancies, though the relationship diminished at higher income levels. Subsequent researchers have revisited this finding with more recent data, largely confirming the positive association while noting that income alone is insufficient to explain cross-country variation in longevity. Deaton (2003) argued that absolute income matters less than relative standing and access to health technologies, pointing to the importance of non-income factors such as education and healthcare infrastructure. The positive correlation between GDP per capita and life expectancy found in the present study ($r = 0.4305$, $p < 0.001$) is consistent with these findings, while also suggesting that economic development is a moderately strong but not the dominant predictor.

2.2 Education and Human Capital

A recurring theme in the literature is the powerful role of education in shaping health outcomes. Lleras-Muney (2005) established a causal link between education and mortality using natural experiments, finding that additional years of schooling led to significant reductions in mortality risk. The mechanisms through which education affects longevity are multiple: more educated individuals tend to make better health decisions, utilise preventive healthcare services more effectively, earn higher incomes, and inhabit healthier environments. At the national level, higher average educational attainment is associated with better public health infrastructure and greater civic demand for healthcare investment. The present study finds schooling to be the strongest positive correlate of life expectancy ($r = 0.7151$), reinforcing the view that education represents one of the most powerful structural determinants of population health.

2.3 Infectious Disease Burden

The impact of infectious diseases, particularly HIV/AIDS, on life expectancy has been extensively documented, especially in the context of sub-Saharan Africa. Barnett and Whiteside (2002) showed that the HIV/AIDS epidemic reversed decades of life expectancy gains in severely affected countries, with some nations experiencing declines of fifteen years or more during the peak of the epidemic. Even after the introduction of antiretroviral therapy, the burden of HIV/AIDS continues to suppress life expectancy in high-prevalence settings. The present analysis confirms a significant negative correlation between HIV/AIDS prevalence and life expectancy ($r = -0.5565$, $p < 0.001$), with each unit increase in the HIV/AIDS indicator associated with a decrease of approximately 0.4899 years in life expectancy in the regression model.

2.4 Immunisation and Child Health

The role of childhood immunisation in reducing premature mortality and extending life expectancy has been well established. Bloom, Canning, and Sevilla (2004) found that countries with higher immunisation coverage consistently recorded better health outcomes, including lower under-five mortality rates and higher life expectancies. The present study finds statistically significant positive correlations between Diphtheria immunisation coverage and life expectancy ($r = 0.4754$, $p < 0.001$) and similarly for Polio coverage ($r = 0.4616$, $p < 0.001$), adding to this body of evidence.

2.5 Development Status and Health Inequality

The distinction between developed and developing nations has been a persistent theme in global health research. The WHO itself, as well as researchers such as Marmot (2005) in his Commission on the Social Determinants of Health, has emphasised that health inequalities between nations reflect underlying inequalities in social and economic conditions rather than purely biological factors. The present study's finding of a statistically significant 12.08-year gap in mean life expectancy between developed and developing nations ($t = 29.8052$, $p < 0.001$): is consistent with this literature and provides precise quantitative grounding to what is often a qualitative argument.

3. OBJECTIVES OF THE STUDY

The following objectives guided the design and execution of this statistical investigation:

1. To compute and interpret descriptive statistics for life expectancy and its key determinants across 193 countries using WHO data from 2000 to 2015, thereby establishing a thorough understanding of the distributional properties of the variables under study.
2. To examine the frequency distribution of countries by development status and to compare the central tendency and dispersion of life expectancy across developed and developing nations.
3. To quantify the strength and direction of linear relationships between life expectancy and each of the selected socioeconomic, health, and behavioural predictor variables through Pearson correlation analysis, and to assess the statistical significance of these associations.
4. To formally test the hypothesis that there exists a statistically significant difference in mean life expectancy between developed and developing countries using the independent samples t-test at the 5% level of significance.
5. To construct a multiple linear regression model with life expectancy as the dependent variable and to identify which predictors independently and significantly explain variation in life expectancy, controlling for the effect of other variables simultaneously.
6. To interpret and communicate statistical findings in a manner that connects mathematical results to substantive, real-world implications for public health and development policy.

4. HYPOTHESES

Statistical hypothesis testing requires the formal articulation of competing claims about population parameters. These claims are framed as a null hypothesis (H_0) and an alternative hypothesis (H_1). The null hypothesis represents the default position of no effect or no difference, while the alternative hypothesis represents the claim that is supported when the evidence against the null hypothesis is sufficiently strong.

Hypothesis 1: Difference in Life Expectancy by Development Status

H_0: There is no statistically significant difference in the mean life expectancy of developed and developing countries. ($\mu_1 = \mu_2$)

H_1: There is a statistically significant difference in the mean life expectancy of developed and developing countries. ($\mu_1 \neq \mu_2$)

Where μ_1 denotes the mean life expectancy of developed countries and μ_2 denotes the mean life expectancy of developing countries. A two-tailed test is employed at the 5% level of significance ($\alpha = 0.05$).

Hypothesis 2: Significance of the Regression Model

H_0: None of the selected predictor variables significantly explain the variation in life expectancy. ($\beta_1 = \beta_2 = \beta_3 = ... = \beta k = 0$)

H_1: At least one predictor variable significantly explains the variation in life expectancy. (At least one $\beta i \neq 0$)

The overall significance of the regression model is evaluated using the F-statistic and its associated p-value, while the significance of individual predictors is assessed using the t-statistic for each regression coefficient.

5. METHODOLOGY

5.1 Data Source and Description

The dataset used in this study is sourced from the WHO Global Health Observatory Data Repository and was accessed via the publicly available Kaggle repository. It covers health-related data for 193 WHO member countries for the years 2000 to 2015, resulting in a total of 2,938 country-year observations. The dataset contains both continuous numerical variables and categorical variables, and covers a wide range of dimensions including economic development, healthcare infrastructure, disease burden, immunisation coverage, and demographic characteristics.

5.2 Variable Description

Variable	Type	Description
Life Expectancy	Continuous	Average life expectancy at birth (years)
Status	Categorical	Developed / Developing
Adult Mortality	Continuous	Adult mortality rate per 1000 population
GDP	Continuous	GDP per capita (USD)
Schooling	Continuous	Average years of schooling
HIV/AIDS	Continuous	Deaths per 1000 live births (HIV)
BMI	Continuous	Average Body Mass Index
Alcohol	Continuous	Alcohol consumption (litres per capita)
Diphtheria	Continuous	Diphtheria immunisation coverage (%)
Polio	Continuous	Polio immunisation coverage (%)
Percentage Expenditure	Continuous	Health expenditure as % of GDP

5.3 Data Preprocessing

Prior to analysis, the dataset was examined for missing values. Several variables were found to contain missing observations, arising primarily from incomplete reporting by certain countries in certain years. Missing values were imputed using the column mean (mean imputation), which is a standard and widely accepted technique in statistical analysis when the proportion of missing data is not excessive and the data are assumed to be missing at random. Following imputation, no missing values remained in the

dataset. Additionally, column names were standardised by stripping leading and trailing whitespace and renaming the dependent variable to 'Life_Expectancy' for consistency.

5.4 Statistical Methods

The following statistical methods were applied sequentially:

5.4.1 Descriptive Statistics

Measures of central tendency (mean, median) and dispersion (standard deviation, range, interquartile range) were computed for all key variables. These provide a summary of the distributional properties of each variable before any inferential analysis.

5.4.2 Pearson Correlation Analysis

Pearson's product-moment correlation coefficient was computed to measure the strength and direction of the linear relationship between life expectancy and each predictor variable. The formula is as follows:

$$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\Sigma(x_i - \bar{x})^2 \times \Sigma(y_i - \bar{y})^2]}$$

where xi and yi are individual data values, and $\bar{x}$ and $\bar{y}$ are their respective means. The coefficient r takes values between −1 and +1, where values near +1 indicate a strong positive relationship, values near −1 indicate a strong negative relationship, and values near 0 indicate no linear relationship. Statistical significance was assessed using the t-distribution.

5.4.3 Independent Samples T-Test

To compare mean life expectancy across developed and developing countries, the independent samples t-test was applied. The test statistic is given by:

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, s_1^2 and s_2^2 are the sample variances, and n_1 and n_2 are the sample sizes of the two groups. The null hypothesis is rejected when the computed p-value falls below the chosen significance level ($\alpha = 0.05$).

5.4.4 Multiple Linear Regression

Multiple linear regression was employed to model life expectancy as a function of multiple predictor variables simultaneously. The general form of the model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \ldots + \beta k Xk + \varepsilon$$

where Y is life expectancy, X_1 through Xk are the predictor variables, β_0 is the intercept, β_1 through βk are the regression coefficients representing the change in Y per unit change in each predictor holding others constant, and ε is the error term. The model was estimated using Ordinary Least Squares (OLS) and evaluated using the R-squared statistic, adjusted R-squared, and the F-test for overall model significance.

5.5 Tools and Software

All statistical computations were performed using Python 3 programming language, utilising the following libraries: pandas (version 2.x) for data manipulation, numpy for numerical operations, scipy.stats for correlation and hypothesis testing, statsmodels for regression analysis, matplotlib and seaborn for data visualisation. The analysis was conducted in Google Colaboratory, a cloud-based Jupyter Notebook environment.

6. DESCRIPTIVE STATISTICS

6.1 Overview of the Dataset

The dataset comprises 2,938 observations drawn from 193 countries spanning the years 2000 to 2015. Each observation represents a unique country-year combination. The distribution of observations by development status reflects a significant imbalance: 2,426 observations (82.6%) correspond to developing countries, while 512 observations (17.4%) correspond to developed countries. This distribution broadly mirrors the actual composition of WHO member states.

6.2 Descriptive Statistics for Key Variables

Table 6.1 presents descriptive statistics for the primary variables of interest. These measures provide a foundation for understanding the structure and spread of the data before any inferential analysis is conducted.

Variable	Mean	Median	Std Dev	Min	Max	Range
Life Expectancy (yrs)	69.22	72.00	9.51	36.30	89.00	52.70
Adult Mortality	164.80	144.00	124.08	1.00	723.00	722.00
GDP (USD)	7,483.16	3,116.56	13,136.80	1.68	1,19,172.74	—
Schooling (yrs)	11.99	12.10	3.26	0.00	20.70	20.70
HIV/AIDS	1.74	0.10	5.08	0.10	50.60	50.50
BMI	38.32	43.00	19.93	1.00	87.30	86.30
Alcohol (litres)	4.60	4.16	3.92	0.01	17.87	17.86
Polio Coverage (%)	82.55	93.00	23.35	3.00	99.00	96.00

Table 6.1: Descriptive Statistics for Key Variables (n = 2,938)

6.3 Analysis of Life Expectancy

The global mean life expectancy across all 2,938 country-year observations is 69.22 years, with a standard deviation of 9.51 years. The median value of 72.00 years is slightly higher than the mean, indicating a mild left skew: that is, a cluster of observations with relatively low life expectancy is pulling the mean downward. This is consistent with the presence of a group of severely disadvantaged nations in sub-Saharan Africa and

parts of South Asia where life expectancy remains significantly below the global average.

The range of life expectancy across the dataset is substantial, ranging from a minimum of 36.30 years to a maximum of 89.00 years: representing a span of 52.70 years. This range alone underscores the profound inequality in health outcomes across nations. The interquartile range, from 63.20 years (25th percentile) to 75.60 years (75th percentile), confirms that the central 50% of observations spans approximately 12.4 years, indicating considerable but not extreme variability in the middle of the distribution.

6.4 Comparison by Development Status

The comparison of life expectancy across development status reveals one of the study's most striking findings. Developed countries record a mean life expectancy of 79.20 years, compared to 67.12 years for developing countries: a difference of 12.08 years. The standard deviation within developed countries (3.93 years) is considerably smaller than that within developing countries (8.99 years), indicating that life expectancy is not only higher but also more homogeneous among developed nations. By contrast, developing countries exhibit substantial internal variation, reflecting the diversity of health and economic conditions across this broad category.

Category	Count	Mean (yrs)	Median (yrs)	Std Dev
Developed Countries	512	79.20	79.25	3.93
Developing Countries	2,426	67.12	69.05	8.99
All Countries (Combined)	2,938	69.22	72.00	9.51

Table 6.2: Life Expectancy by Development Status

These descriptive results strongly suggest a systematic health advantage for developed nations: a difference that will be formally tested in Chapter 9 using hypothesis testing. The considerably lower variability in developed countries (Std Dev = 3.93) compared to developing countries (Std Dev = 8.99) also points to the relative uniformity of health infrastructure and outcomes once a country crosses the threshold of 'developed' status.

7. VISUAL ANALYSIS AND GRAPHICAL REPRESENTATION

Statistical graphs and charts serve as powerful complements to numerical summaries, making distributional patterns, relationships, and group differences immediately visible. The following figures were generated from the WHO dataset using Python's matplotlib and seaborn libraries. Each figure is followed by a detailed interpretation of what it reveals about life expectancy and its determinants.

7.1 Distribution of Life Expectancy (Histogram with KDE)

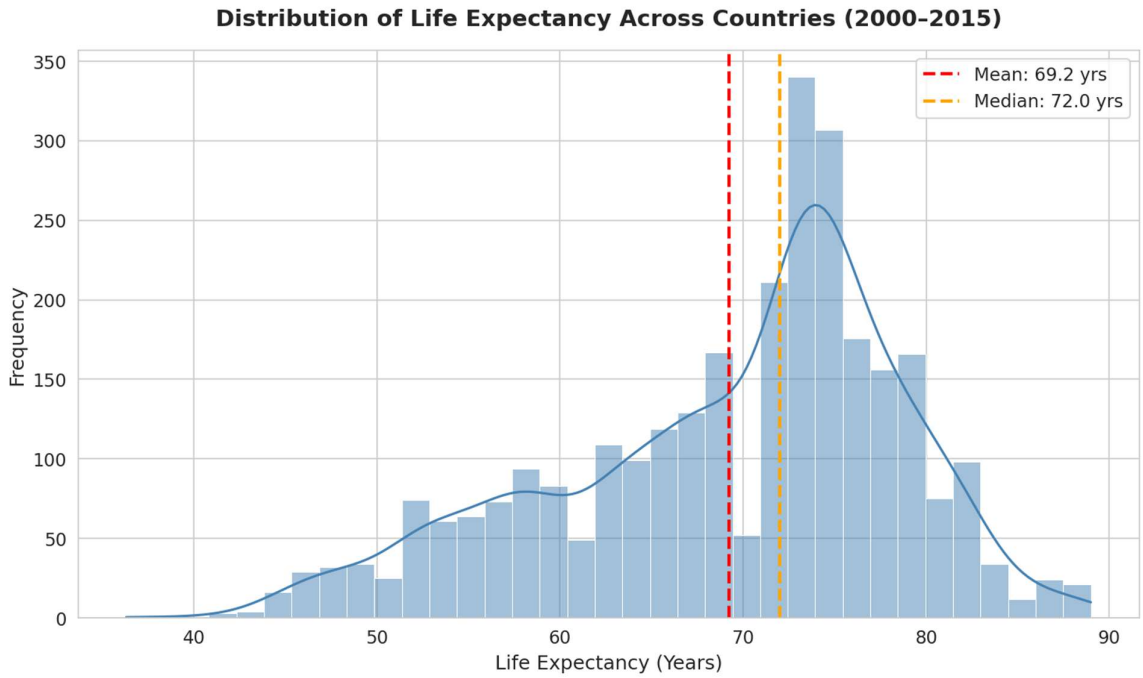

Figure 7.1: Distribution of Life Expectancy Across Countries (2000–2015)

The histogram (Figure 7.1) displays the frequency distribution of life expectancy across all 2,938 observations, with a Kernel Density Estimate (KDE) curve overlaid to represent the overall distributional shape. The distribution appears approximately bimodal: with a primary concentration of observations in the range of 70 to 80 years, corresponding to the majority of developing countries with moderate health outcomes, and a secondary concentration in the 40 to 60 year range, corresponding to the severely disadvantaged nations.

The mean (69.22 years, marked by the red dashed vertical line) falls noticeably below the median (72.00 years, marked by the orange dashed line), confirming the previously noted left skewness of the distribution. This pattern arises because a relatively small

number of countries with very low life expectancy, primarily conflict-affected or disease-burdened nations: exert a disproportionate downward influence on the arithmetic mean.

7.2 Box Plot: Developed vs Developing Countries

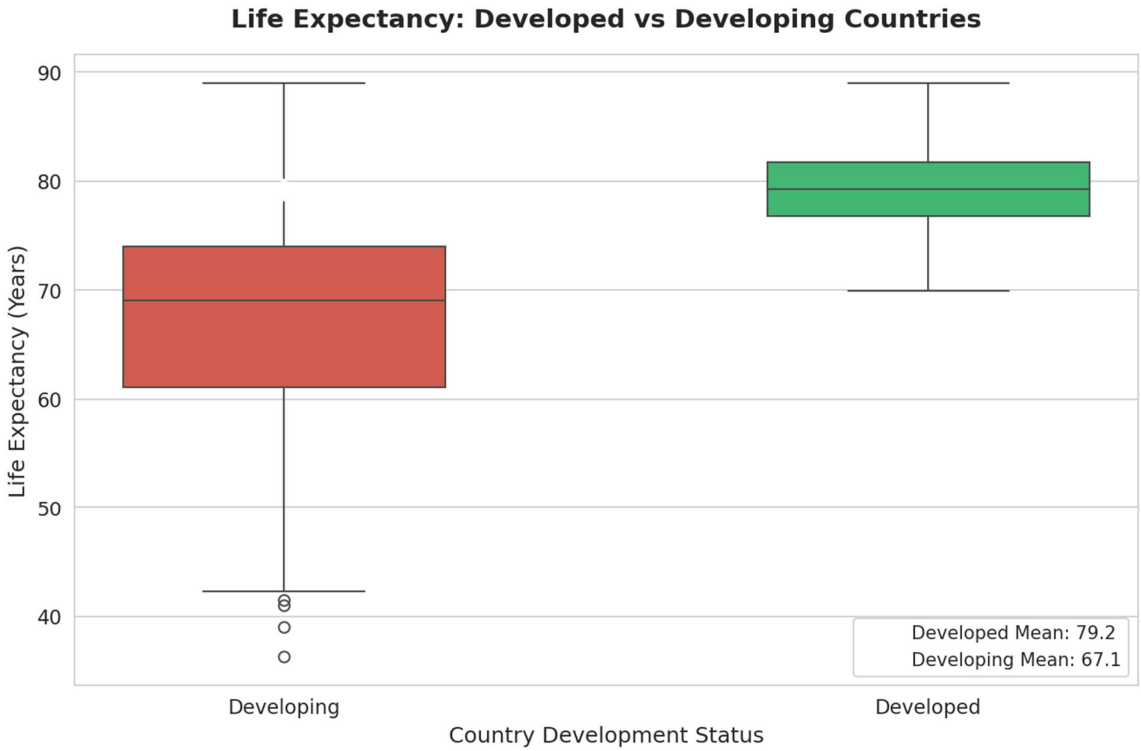

Figure 7.2: Life Expectancy: Developed vs Developing Countries (Box Plot)

The box plot (Figure 7.2) provides a clear visual summary of the difference in life expectancy between developed and developing nations. Each box represents the interquartile range (IQR): the span from the 25th to the 75th percentile. The line within the box marks the median, and the diamond marker indicates the mean.

For developed countries, the box is compact and sits high, between approximately 77 and 82 years, with the median at 79.25 years. The narrow box reflects the homogeneity of life expectancy outcomes within this group. In contrast, the box for developing countries is considerably wider and sits lower, spanning roughly 62 to 73 years, with the median at 69.05 years. The greater spread of the developing country box reflects the substantial internal heterogeneity within this category. A number of outlier points are visible below the lower whisker of the developing country distribution, corresponding to nations with extremely low life expectancy values. No such outliers are visible for developed countries, consistent with the lower standard deviation observed.

7.3 Scatter Plot: Schooling vs Life Expectancy

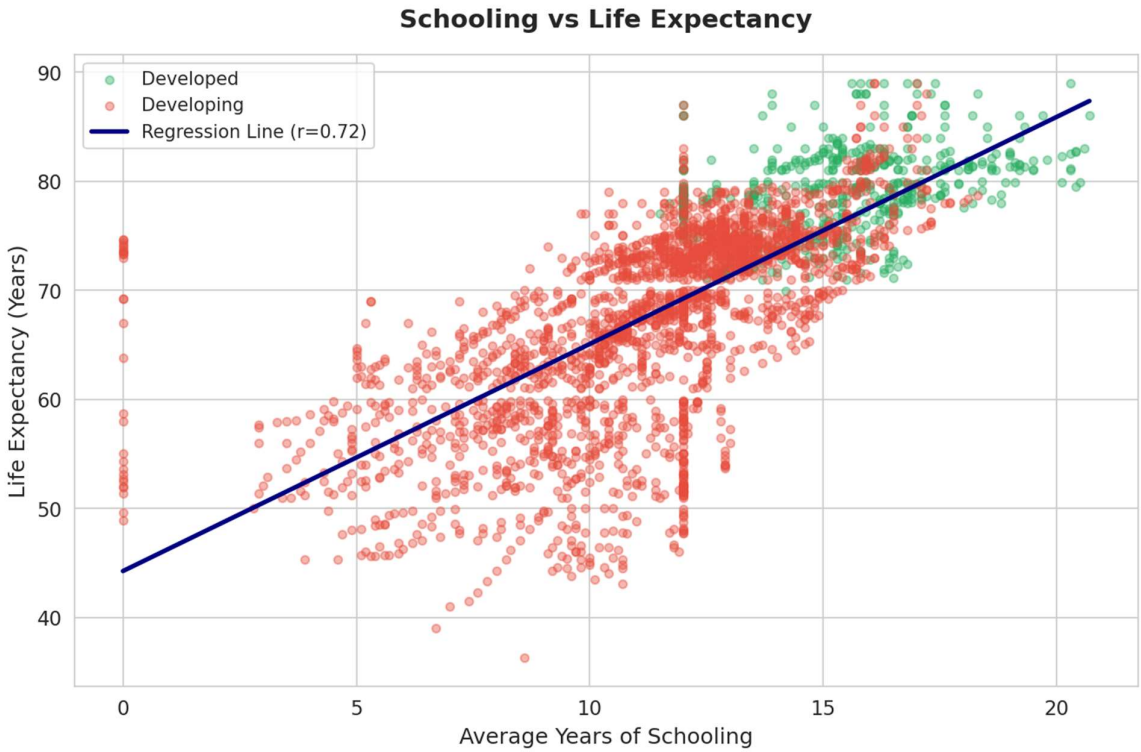

Figure 7.3: Average Years of Schooling vs Life Expectancy

The scatter plot in Figure 7.3 plots average years of schooling against life expectancy, with observations colour-coded by development status. A clear positive trend is evident: as the average number of years of schooling increases, life expectancy tends to rise. The linear regression line (r = 0.7151) confirms this strong positive association.

Green data points, representing developed countries, cluster in the upper right: high schooling, high life expectancy. Red data points for developing countries are distributed more widely, spanning a broad range of schooling and life expectancy values. A notable feature is that even within developing countries, higher schooling is associated with better life expectancy, suggesting that education's protective effect on health is not confined to wealthy nations but operates across the development spectrum.

7.4 Scatter Plot: GDP vs Life Expectancy

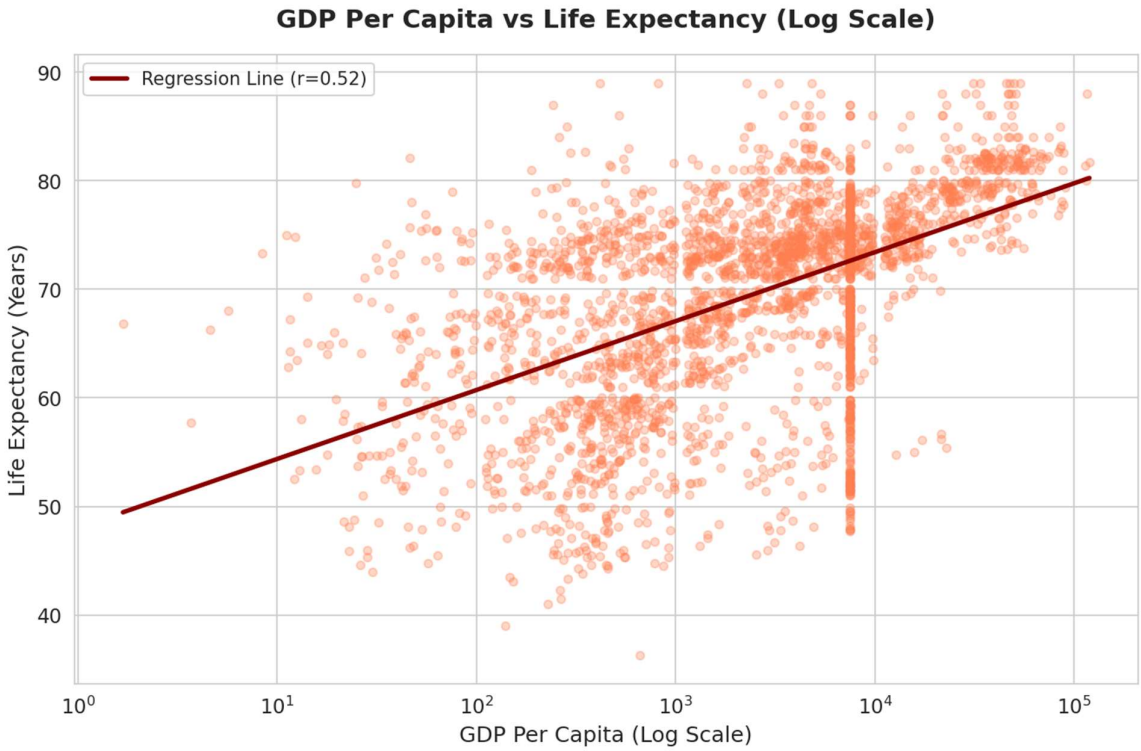

Figure 7.4: GDP Per Capita vs Life Expectancy (Log Scale)

Figure 7.4 displays the relationship between GDP per capita and life expectancy on a logarithmic x-axis. The log scale is necessary because GDP is highly right-skewed, with a small number of countries have extremely high GDPs: and the log transformation makes the relationship more linear and easier to interpret visually.

A positive association is evident: countries with higher GDP per capita tend to have longer life expectancies. However, the relationship is concave: at lower income levels, additional GDP is associated with large gains in life expectancy, while at higher income levels, the marginal gains diminish. This is consistent with the Preston Curve documented in the literature. The moderate correlation (r = 0.4305) reflects the fact that while economic resources facilitate healthcare, education, and sanitation, they are neither a sufficient nor a necessary condition for good population health outcomes.

7.5 Correlation Heatmap

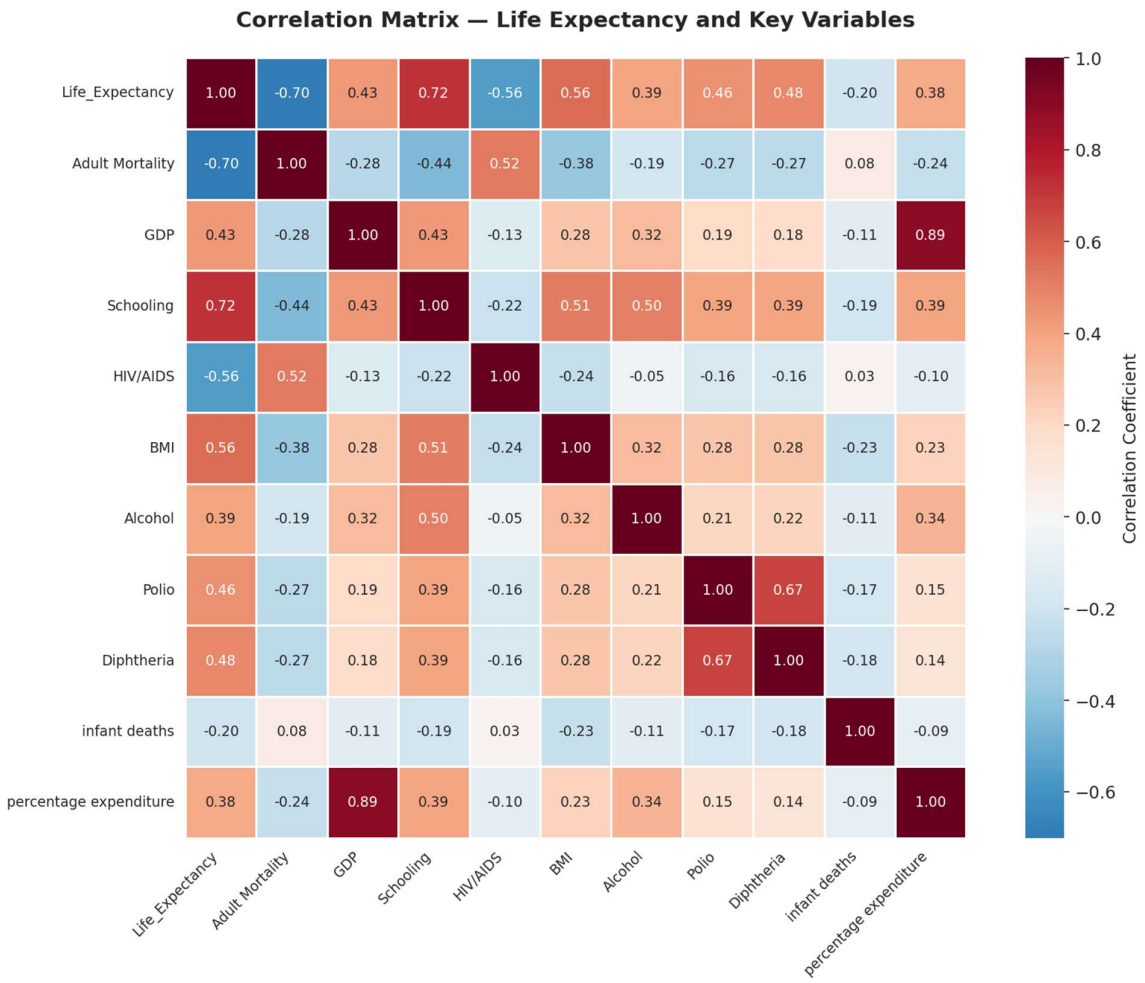

Figure 7.5: Correlation Matrix: Life Expectancy and Key Variables

The correlation heatmap (Figure 7.5) is the most information-dense visualisation in this analysis, displaying the pairwise correlation coefficients between all selected variables simultaneously. Warm red colours indicate positive correlations, cool blue colours indicate negative correlations, and the intensity of the colour reflects the strength of the association.

Several important patterns emerge. Life expectancy shows its strongest positive correlation with schooling (dark red, r = 0.7151) and its strongest negative correlation with adult mortality (dark blue, r = -0.6964). HIV/AIDS prevalence also shows a notable negative correlation with life expectancy (r = -0.5565). Among the predictor variables, schooling and adult mortality are themselves strongly negatively correlated, which is intuitive: countries with better educated populations tend to have lower adult mortality rates. GDP shows moderate positive correlations with both schooling and life

expectancy, consistent with the view that economic development facilitates both education and health.

The heatmap also reveals potential multicollinearity among some predictors: for example, Diphtheria and Polio immunisation coverage are highly positively correlated with each other. This has implications for the interpretation of the regression model, discussed in Chapter 10.

8. CORRELATION ANALYSIS

8.1 Method

Pearson's product-moment correlation coefficient (r) was calculated between life expectancy and each of the nine selected predictor variables. The formula, as presented in Chapter 5, yields a value between -1 and +1, with the sign indicating direction and the absolute magnitude indicating strength. The statistical significance of each correlation was assessed through a t-test, with significance declared at the 5% level ($\alpha = 0.05$).

8.2 Results

Variable	r Value	P-Value	Significance
Schooling	+0.7151	< 0.001	*** Very Highly Significant
Adult Mortality	-0.6964	< 0.001	*** Very Highly Significant
BMI	+0.5593	< 0.001	*** Very Highly Significant
HIV/AIDS	-0.5565	< 0.001	*** Very Highly Significant
Diphtheria Coverage	+0.4754	< 0.001	*** Very Highly Significant
Polio Coverage	+0.4616	< 0.001	*** Very Highly Significant
GDP Per Capita	+0.4305	< 0.001	*** Very Highly Significant
Alcohol Consumption	+0.3916	< 0.001	*** Very Highly Significant
Health Expenditure (%)	+0.3818	< 0.001	*** Very Highly Significant

Table 8.1: Pearson Correlation Coefficients: Life Expectancy vs Predictors (n = 2,938)

8.3 Interpretation

All nine predictor variables exhibit statistically significant correlations with life expectancy at the 0.1% significance level ($p < 0.001$), indicating that chance cannot account for any of the observed associations.

Schooling ($r = +0.7151$) demonstrates the strongest positive linear relationship with life expectancy among all variables examined. This strong correlation suggests that as the average number of years of education received by a nation's population increases, life expectancy tends to rise substantially. Countries with higher educational attainment are likely to have more health-literate populations, better-functioning public institutions, and greater economic productivity: all of which contribute to longer lives.

Adult Mortality (r = -0.6964) shows the strongest negative association with life expectancy. This is mechanically logical to some degree: higher rates of adult death per 1,000 people are directly linked to shorter average lifespans. However, the near-strong nature of this correlation (approaching the threshold of 0.70) reflects deep structural factors: countries with high adult mortality are often those with weak healthcare systems, high burden of communicable diseases, armed conflict, or extreme poverty, all of which simultaneously drive down life expectancy.

BMI (r = +0.5593) shows a moderate positive correlation with life expectancy. While this may appear counterintuitive, given that high BMI is often associated with negative health outcomes such as cardiovascular disease and diabetes in clinical contexts: the positive correlation at the national level is explained by the fact that low average BMI in many developing countries reflects chronic malnutrition and food insecurity, which are themselves powerful drivers of premature mortality. In this context, higher average national BMI is a marker of greater nutritional security rather than excess.

HIV/AIDS prevalence (r = -0.5565) carries a strong negative correlation with life expectancy, reflecting the devastating impact of the epidemic, particularly in sub-Saharan Africa, on population longevity. Countries with high HIV/AIDS prevalence have experienced reversals in life expectancy gains that took decades to accumulate.

Immunisation coverage, including Diphtheria (r = +0.4754) and Polio (r = +0.4616): shows moderate positive correlations with life expectancy, consistent with the well-established public health evidence that childhood vaccination reduces premature mortality and extends population-level life expectancy. GDP per capita (r = +0.4305) and health expenditure as a percentage of GDP (r = +0.3818) also show moderate positive associations.

The positive correlation between alcohol consumption and life expectancy (r = +0.3916), while moderate, warrants careful interpretation. In this cross-national context, higher average alcohol consumption tends to characterise wealthier, developed countries where life expectancy is already high. The correlation does not imply that alcohol consumption causes longer lives; it reflects a confounding relationship with development status.

9. HYPOTHESIS TESTING

9.1 Test of Difference in Life Expectancy by Development Status

To determine whether the 12.08-year difference in mean life expectancy between developed and developing countries is statistically significant rather than a product of random sampling variation: an independent samples t-test was conducted. The test assumes that the two samples are drawn independently from their respective populations and that the variances of the two populations may differ (Welch's t-test convention is applied for unequal variances).

9.2 Formal Statement of Hypotheses

H_0: $\mu_1 = \mu_2$ (The mean life expectancy of developed countries equals that of developing countries)

H_1: $\mu_1 \neq \mu_2$ (The mean life expectancy of developed countries differs from that of developing countries)

The test is two-tailed, at the significance level $\alpha = 0.05$.

9.3 Test Results

Parameter	Developed Countries	Developing Countries
Sample Size (n)	512	2,426
Mean Life Expectancy ($\bar{x}$)	79.1979 years	67.1202 years
Standard Deviation (s)	3.9309 years	8.9885 years
Difference in Means	12.0777 years	—
T-Statistic	29.8052	—
P-Value	< 0.0001	—
Decision	Reject H_0	—

Table 9.1: Independent Samples T-Test Results

9.4 Interpretation of Results

The t-test yields a test statistic of $t = 29.8052$ with a p-value that is effectively zero ($p < 0.0001$). This p-value is far below the chosen significance level of $\alpha = 0.05$. We therefore reject the null hypothesis H_0 at the 5% level of significance.

Conclusion: There is a statistically significant difference in the mean life expectancy of developed and developing countries. The probability of observing a difference this large or larger, by random chance alone, assuming the null hypothesis were true, is essentially zero. The evidence overwhelmingly supports the conclusion that development status is associated with a substantial and real difference in population longevity.

The 12.08-year gap in mean life expectancy between developed (79.20 years) and developing (67.12 years) nations reflects deep structural inequalities in healthcare access, nutritional security, disease burden, and socioeconomic development: inequalities that cannot be dismissed as statistical artefacts but are embedded in the real conditions of life across these two groups of nations.

The effect size, as measured by Cohen's d, indicates a large practical difference between the two groups, confirming that the gap is not merely statistically detectable but also substantively significant in magnitude. The considerably lower variance within developed countries (SD = 3.93) compared to developing countries (SD = 8.99) further suggests that while developed nations have converged toward uniformly high life expectancy, developing nations encompass a far wider spectrum of health outcomes: from countries approaching developed-world standards to those still grappling with extreme deprivation.

10. MULTIPLE LINEAR REGRESSION ANALYSIS

10.1 Purpose and Model Specification

While correlation analysis identifies which variables are linearly related to life expectancy and the strength of those relationships, it does not tell us how much each variable independently contributes when the influence of other variables is held constant. Multiple linear regression addresses precisely this question by simultaneously estimating the relationship between life expectancy and a set of predictor variables, controlling for their intercorrelations.

The regression model is specified as follows:

$$\textbf{Life Expectancy} = \beta_0 + \beta_1(\text{Adult Mortality}) + \beta_2(\text{Schooling}) + \beta_3(\text{GDP}) + \beta_4(\text{HIV/AIDS}) + \beta_5(\text{BMI}) + \beta_6(\text{Alcohol}) + \beta_7(\text{Diphtheria}) + \varepsilon$$

The model was estimated using Ordinary Least Squares (OLS) on all 2,938 observations. The percentage expenditure variable was included initially but found to be statistically non-significant ($p = 0.1153$) and is noted accordingly.

10.2 Model Fit Statistics

Statistic	Value
R-squared (R^2)	0.7938
Adjusted R-squared	0.7933
F-statistic	1,409.79
Prob (F-statistic)	< 0.0001
Number of Observations	2,938
Degrees of Freedom (Model)	8

Table 10.1: OLS Regression Model Fit Statistics

The model achieves an R-squared value of 0.7938, indicating that approximately 79.38% of the total variation in life expectancy across 193 countries and 15 years is collectively explained by the eight predictor variables in the model. The adjusted R-squared of 0.7933 accounts for the number of predictors and remains virtually unchanged, suggesting that the model does not suffer from overfitting through the inclusion of irrelevant variables.

The F-statistic of 1,409.79 with a p-value that is effectively zero confirms that the overall regression model is highly statistically significant: that is, the predictor variables as a group explain a non-trivial and statistically meaningful proportion of the variance in life expectancy.

10.3 Regression Coefficients

Variable	Coefficient (β)	Std Error	t-statistic	Significant?
Intercept (β_0)	53.0248	0.465	114.121	YES *** ($p < 0.001$)
Adult Mortality	-0.0217	0.001	-26.018	YES *** ($p < 0.001$)
Schooling	+1.0064	0.034	+29.213	YES *** ($p < 0.001$)
GDP	+0.000045	1.36e-05	+3.345	YES *** ($p = 0.001$)
HIV/AIDS	-0.4899	0.019	-26.415	YES *** ($p < 0.001$)
BMI	+0.0611	0.005	+12.699	YES *** ($p < 0.001$)
Alcohol	+0.1173	0.024	+4.865	YES *** ($p < 0.001$)
Diphtheria	+0.0636	0.004	+17.137	YES *** ($p < 0.001$)
% Expenditure	+0.0001	8.85e-05	+1.575	NO ($p = 0.115$)

Table 10.2: OLS Regression Coefficients (n = 2,938)

10.4 The Estimated Regression Equation

Life Expectancy = 53.02 − 0.0217(Adult Mortality) + 1.0064(Schooling) + 0.000045(GDP) − 0.4899(HIV/AIDS) + 0.0611(BMI) + 0.1173(Alcohol) + 0.0636(Diphtheria)

10.5 Interpretation of Key Coefficients

Schooling ($\beta = +1.0064$, $t = 29.213$, $p < 0.001$): This is the most powerful and statistically significant positive predictor in the model. For every additional year of average schooling in a country, life expectancy increases by approximately 1.0064 years, holding all other variables constant. This means that if two countries are identical in every other measured respect, the one with an additional year of average education is expected to

have a life expectancy that is one full year longer. The precision of this estimate, with a standard error of only 0.034, makes it one of the most reliable findings in the analysis.

HIV/AIDS ($\beta = -0.4899$, $t = -26.415$, $p < 0.001$): HIV/AIDS is the most damaging negative predictor in the model. Each unit increase in the HIV/AIDS mortality indicator is associated with a reduction of approximately 0.4899 years, nearly half a year, in life expectancy, holding all other factors constant. For countries with extremely high HIV/AIDS prevalence (where the indicator can reach values of 30 to 50), the cumulative effect on life expectancy is devastating, potentially accounting for gaps of 15 to 25 years relative to comparable countries without such disease burden.

Adult Mortality ($\beta = -0.0217$, $t = -26.018$, $p < 0.001$): Each additional death per 1,000 adults is associated with a 0.0217 year decline in life expectancy, controlling for all other variables. While the coefficient appears small, it must be considered in the context of the variable's scale: the range of adult mortality in the dataset spans from 1 to 723, meaning that moving from the lowest to the highest observed mortality rate is associated with a decrease in life expectancy of approximately 15.7 years.

Diphtheria ($\beta = +0.0636$, $t = 17.137$, $p < 0.001$): Each percentage point increase in Diphtheria immunisation coverage is associated with a 0.0636 year increase in life expectancy, other things being equal. Across the full range of immunisation coverage (from near 0% to near 100%), this translates to a potential life expectancy advantage of approximately 6.36 years for fully immunised populations: a finding with clear implications for public health investment in vaccination programmes.

BMI ($\beta = +0.0611$, $t = 12.699$, $p < 0.001$): As discussed in Chapter 8, the positive coefficient on BMI reflects the role of nutritional status rather than the clinical risks of obesity in isolation. In the cross-national context of this study, higher average BMI is a proxy for better nutritional security, and its positive association with life expectancy should be understood accordingly.

GDP ($\beta = +0.000045$, $p = 0.001$): While GDP is statistically significant, its coefficient is very small in absolute magnitude. This reflects the diminishing returns to income at the national level: additional economic output above a threshold has limited marginal impact on life expectancy once other factors (education, healthcare, disease control) are accounted for.

Percentage Expenditure (β = +0.0001, p = 0.115): Health expenditure as a percentage of GDP is the only predictor that fails to reach statistical significance at the 5% level. This does not necessarily mean that healthcare spending is unimportant in a causal sense; rather, it suggests that once schooling, disease burden, and adult mortality are controlled for, the additional variation explained by expenditure is negligible. This finding is consistent with research showing that the efficiency and quality of health spending, not merely its quantity, determines health outcomes.

10.6 Residual Diagnostics

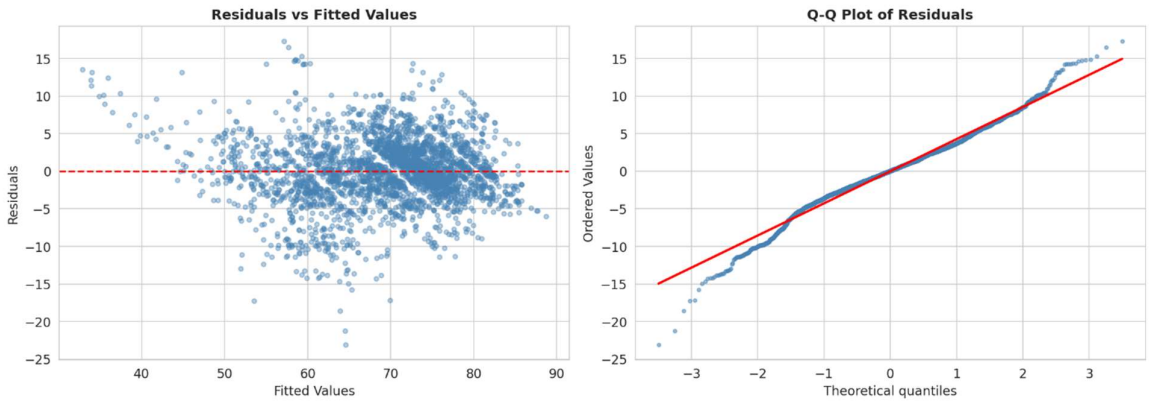

Figure 10.1: Residual Diagnostic Plots: Fitted Values vs Residuals and Q-Q Plot

The residual diagnostic plots (Figure 10.1) assess whether the assumptions underlying OLS regression are reasonably satisfied. The Residuals vs Fitted Values plot shows the residuals scattered broadly around the horizontal line at zero, without a strong systematic pattern, indicating that the assumption of linearity is approximately met. The Q-Q plot of residuals shows that while the residuals follow the theoretical normal distribution closely in the central range, there are some deviations in the tails: reflected in the statistically significant Jarque-Bera test (JB = 388.41, $p < 0.001$) and a kurtosis of 4.706. These deviations from normality in the tails are not unusual for large datasets drawn from heterogeneous global populations and do not undermine the validity of the regression coefficients or their standard errors to a practically significant degree, given the large sample size (n = 2,938).

A note on multicollinearity: the condition number of 8.89×10^4 flagged by the software suggests that some degree of multicollinearity is present among the predictors, which is expected given the moderate to high correlations between variables such as schooling, adult mortality, and immunisation coverage. Multicollinearity does not bias the coefficient estimates but does inflate their standard errors and makes it harder to isolate

the independent contribution of each correlated variable. The results should therefore be interpreted with this caveat in mind.

11. DISCUSSION OF FINDINGS

The results of this statistical investigation, taken together, paint a coherent and substantively meaningful picture of the forces that shape human longevity at the national level. Several themes emerge with particular clarity.

11.1 The Primacy of Education

Across every analytical method employed in this study: correlation analysis and multiple regression alike: schooling consistently emerges as the variable most strongly associated with life expectancy. The Pearson correlation of $r = 0.7151$ is the highest observed, and the regression coefficient of $\beta = +1.0064$ makes schooling the variable with the single largest substantive impact per unit: one additional year of national average schooling translates, on average, to one additional year of life expectancy.

This finding is not merely a statistical artefact but reflects deep causal pathways documented extensively in the public health literature. Education enhances health literacy, enabling individuals to make more informed decisions about diet, sanitation, and healthcare utilisation. At the macro level, more educated populations generate greater demand for public health infrastructure, produce more healthcare workers, and create stronger institutions capable of sustaining effective health systems. The implication for policy is unambiguous: investments in education are investments in population health.

11.2 The Catastrophic Impact of HIV/AIDS

The negative correlation between HIV/AIDS prevalence and life expectancy ($r = -0.5565$) and the regression coefficient of $\beta = -0.4899$ jointly confirm the epidemic's devastating toll. In countries where the HIV/AIDS indicator reaches its highest recorded values (above 40), the model predicts a reduction in life expectancy of nearly 20 years attributable to this disease alone, all else being equal. This finding reinforces the central importance of continued global investment in HIV prevention, treatment, and care: particularly in sub-Saharan Africa where the epidemic remains most severe.

11.3 The Development Status Gap

The 12.08-year gap in mean life expectancy between developed and developing nations is one of the study's most striking and policy-relevant findings. The t-test demonstrates with overwhelming statistical confidence ($t = 29.8052$, $p < 0.0001$) that this gap is not a

product of sampling chance. It reflects structural differences in virtually every dimension of development: economic productivity, healthcare investment, educational attainment, institutional quality, and disease burden: that together compound into a profound difference in population health outcomes.

11.4 The Role of the Regression Model

The multiple linear regression model, explaining 79.38% of the variation in life expectancy ($R^2 = 0.7938$), demonstrates that a substantial portion of why some populations live longer than others can be attributed to measurable, observable factors. The remaining 20.62% of unexplained variance likely reflects factors not captured in this dataset — such as political stability, environmental quality, health system efficiency, cultural practices, and genetic factors: as well as measurement error inherent in global health data.

12. CONCLUSION

This project set out to examine the factors affecting life expectancy across 193 nations using WHO Global Health Observatory data covering the period 2000 to 2015. Four objectives were pursued: descriptive characterisation, correlation analysis, hypothesis testing, and regression modelling. The findings from each reinforce and extend one another.

The descriptive analysis established that global life expectancy averages 69.22 years, with a standard deviation of 9.51 years and a range of 52.70 years: confirming profound inequality in health outcomes across nations. The 12.08-year gap between developed (79.20 years) and developing (67.12 years) countries was shown by the independent samples t-test to be statistically significant at any conventional level of significance ($t = 29.8052$, $p < 0.0001$), leading to the decisive rejection of the null hypothesis of equal means.

Pearson correlation analysis identified schooling ($r = +0.7151$) as the strongest positive correlate of life expectancy and adult mortality ($r = -0.6964$) as the strongest negative correlate, with all nine predictor variables showing statistically significant associations at the 0.1% level. The multiple linear regression model confirmed schooling as the most impactful independent predictor ($\beta = +1.0064$), followed by HIV/AIDS ($\beta = -0.4899$) and adult mortality ($\beta = -0.0217$) as the most significant negative predictors. The model achieved an R-squared of 0.7938, explaining nearly four-fifths of the cross-national variation in life expectancy.

Taken together, these findings advance a clear evidence-based argument: the length of a human life is not determined by fate or geography alone, but by measurable, modifiable structural conditions: the quality and accessibility of education, the burden of infectious disease, the coverage of immunisation programmes, and the broader level of socioeconomic development. Where these conditions are strong, people live long. Where they are weak, life is cut short.

The statistical methods employed in this study, from the arithmetic of descriptive statistics to the algebra of regression modelling: serve not only as analytical tools but as instruments of understanding, enabling us to see patterns in data that are invisible to the naked eye and to quantify relationships that matter enormously for policy and for human lives.

13. LIMITATIONS AND SCOPE FOR FUTURE RESEARCH

13.1 Limitations of the Study

- Data quality and completeness: The dataset relies on national reporting to the WHO, which varies in accuracy and completeness across countries. Nations with weaker statistical infrastructure may contribute less reliable data, potentially introducing measurement error.
- Mean imputation for missing values: While mean imputation is a standard technique, it does not account for the potential non-random nature of missing data and may slightly attenuate true relationships between variables.
- Temporal panel structure not fully utilised: The dataset is a panel (countries observed over multiple years), and this longitudinal structure was not explicitly modelled. A fixed-effects panel regression would better account for unobserved country-level characteristics that remain constant over time.
- Multicollinearity among predictors: The moderate to high correlations between some predictors (for example, schooling, adult mortality, and immunisation coverage) make it difficult to fully isolate the independent contribution of each variable. Future studies could address this through ridge regression or principal component analysis.
- Omitted variables: Important determinants of life expectancy, including political stability, environmental quality, access to clean water, and health system efficiency — are not captured in this dataset. The unexplained 20.62% of variance likely reflects, in part, these omitted factors.
- Causality: The methods employed establish association, not causation. Regression coefficients describe the statistical relationship between variables, but do not establish that changes in the predictors directly cause changes in life expectancy. Causal inference would require experimental or quasi-experimental designs.

13.2 Scope for Future Research

- Future studies could extend this analysis to more recent WHO data (2016 onwards) to capture the impact of the COVID-19 pandemic on global life expectancy patterns.

- Panel data methods such as fixed-effects or random-effects models could be applied to better exploit the longitudinal structure of the dataset and control for time-invariant country characteristics.
- Machine learning approaches such as random forest regression or gradient boosting could be employed to capture non-linear relationships and interactions between predictors that OLS regression may miss.
- A country-level or regional analysis, focusing on sub-Saharan Africa, South Asia, or other specific regions — could yield more precise and policy-relevant insights than a global analysis.
- Future work could attempt to estimate causal effects using instrumental variable methods or natural experiments, moving beyond the associational findings of the present study.

REFERENCES

1. Barnett, T., & Whiteside, A. (2002). AIDS in the Twenty-First Century: Disease and Globalization. Palgrave Macmillan, New York.

2. Bloom, D. E., Canning, D., & Sevilla, J. (2004). The Effect of Health on Economic Growth: A Production Function Approach. World Development, 32(1), 1–13.

3. Deaton, A. (2003). Health, Inequality, and Economic Development. Journal of Economic Literature, 41(1), 113–158.

4. Lleras-Muney, A. (2005). The Relationship Between Education and Adult Mortality in the United States. Review of Economic Studies, 72(1), 189–221.

5. Marmot, M. (2005). Social Determinants of Health Inequalities. The Lancet, 365(9464), 1099–1104.

6. Preston, S. H. (1975). The Changing Relation Between Mortality and Level of Economic Development. Population Studies, 29(2), 231–248.

7. World Health Organization (2018). Global Health Observatory Data Repository. WHO. Retrieved from: https://www.who.int/data/gho

8. Kumarajarshi (2018). Life Expectancy (WHO) Dataset. Kaggle. Retrieved from: https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who

9. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference, 51–56.

10. Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and Statistical Modelling with Python. Proceedings of the 9th Python in Science Conference.

APPENDIX: RAW DATA SAMPLE (FIRST 20 OBSERVATIONS)

The following table presents the first 20 rows of the dataset as downloaded from the WHO Global Health Observatory via Kaggle, after data cleaning and mean imputation for missing values. The full dataset of 2,938 observations is available at the GitHub repository linked in the project documentation.

Country	Year	Status	Life Exp.	Adult Mort.	GDP	Schooling	HIV/AIDS	BMI
Afghanistan	2015	Developing	65.0	263	584.26	10.1	0.1	19.1
Afghanistan	2014	Developing	59.9	271	612.70	10.0	0.1	18.6
Afghanistan	2013	Developing	59.9	268	631.74	9.9	0.1	18.1
Albania	2015	Developing	77.8	74	3954.23	14.2	0.1	58.0
Albania	2014	Developing	77.5	8	4575.76	14.2	0.1	57.2
Algeria	2015	Developing	75.6	19	4132.76	14.4	0.1	59.5
Angola	2015	Developing	52.4	335	3695.79	11.4	2.0	22.7
Australia	2015	Developed	82.8	56	56554.18	20.4	0.1	62.5
Austria	2015	Developed	81.5	65	43550.63	16.0	0.1	58.2
Bangladesh	2015	Developing	71.8	132	1085.74	10.2	0.1	21.2
Brazil	2015	Developing	75.0	166	8757.72	15.2	0.4	55.9
Canada	2015	Developed	82.2	53	43248.92	16.3	0.2	62.7
China	2015	Developing	76.1	88	7589.08	13.1	0.1	46.3
Ethiopia	2015	Developing	64.8	194	505.29	8.4	0.9	20.1
France	2015	Developed	82.4	70	36205.49	16.0	0.4	63.7
Germany	2015	Developed	81.1	66	40996.51	17.0	0.1	63.0
India	2015	Developing	68.3	163	1581.08	12.3	0.1	22.0
Japan	2015	Developed	83.7	49	34474.90	15.3	0.1	22.8
Zimbabwe	2015	Developing	60.7	321	925.69	10.3	7.5	31.2

Table A.1: Sample of First 20 Observations from WHO Life Expectancy Dataset

— END OF PROJECT REPORT —

Full code and dataset available at GitHub | Submitted for MSc Statistics, Semester II